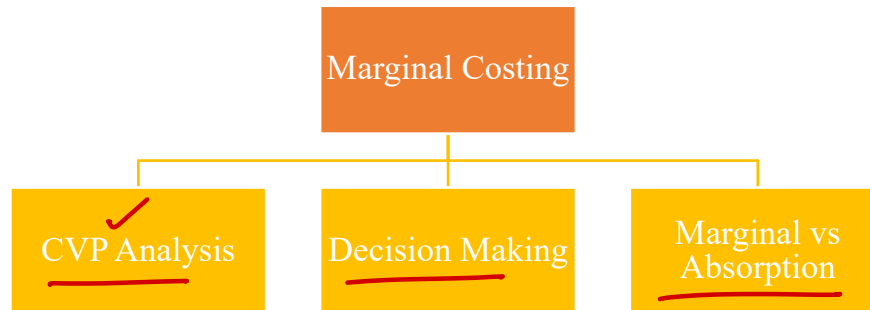


MARGINAL COSTING

1. Marginal Costing



Sales	→	XX
<u>(-) Variable cost</u>	→	<u>XX</u>
Contribution	→	XX
<u>(-) Fixed Cost</u>	→	<u>XX</u>
<u>Profit</u>	→	<u>XX</u>

2. Contribution = Sales – Variable Cost = Fixed Cost + Profit

3. Profit Volume (PV) Ratio or Contribution Ratio

- This ratio doesn't change with change in level of output
- This ratio changes with change in either selling price per unit or variable cost per unit

$$\text{PV Ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100 = \frac{\text{Sales} - \text{Variable Costs}}{\text{Sales}} \times 100 = \frac{\text{Fixed cost} + \text{Profit}}{\text{Sales}} \times 100$$

$$\text{PV Ratio} = 100 - \text{Variable Cost Ratio}$$

$$\text{PV Ratio} = \frac{\text{Change in Contribution}}{\text{Change in Sales}} \times 100 = \frac{\text{Change in Profit}}{\text{Change in Sales}} \times 100$$

4. Break-even point (BEP)

- It is the level of sales at which there is neither any profit nor any sales
- In other words, it is the level of sales at which contribution is just able to recovery FC.

$$\text{Break-even Point (units of sale)} = \frac{\text{Fixed Cost}}{\text{Contribution per unit}} \quad \checkmark$$

$$\begin{aligned} \text{Break-even Point (in sales value)} &= \frac{\text{Fixed Cost}}{\text{P/V Ratio}} \quad \checkmark \\ &= \text{Break-even point units} \times \text{Selling price per unit} \end{aligned}$$

5. Cash Break-even Point

➤ It is level of sales at which cash profit or loss is zero.

$$\circ \text{ Cash Break-even Point (units of sale)} = \frac{\text{Cash Fixed Cost}}{\text{Contribution per unit}}$$

$$\circ \text{ Cash Break-even Point (in sales value)} = \frac{\text{Cash Fixed Cost}}{\text{P/V Ratio}}$$

6. Required sales for a given level of profit

$$\text{Sales to earn desired profit (units)} = \frac{\text{Fixed cost} + \text{Desired profit}}{\text{Contribution per unit}}$$

$$\text{Sales to earn desired profit (in ₹)} = \frac{\text{Fixed cost} + \text{Desired profit}}{\text{P/V Ratio}}$$

7. Margin of Safety (MOS)

It is the level of sales over and above break-even sales

$$\text{Margin of Safety (in ₹)} = \text{Actual sales} - \text{Break-even sales}$$

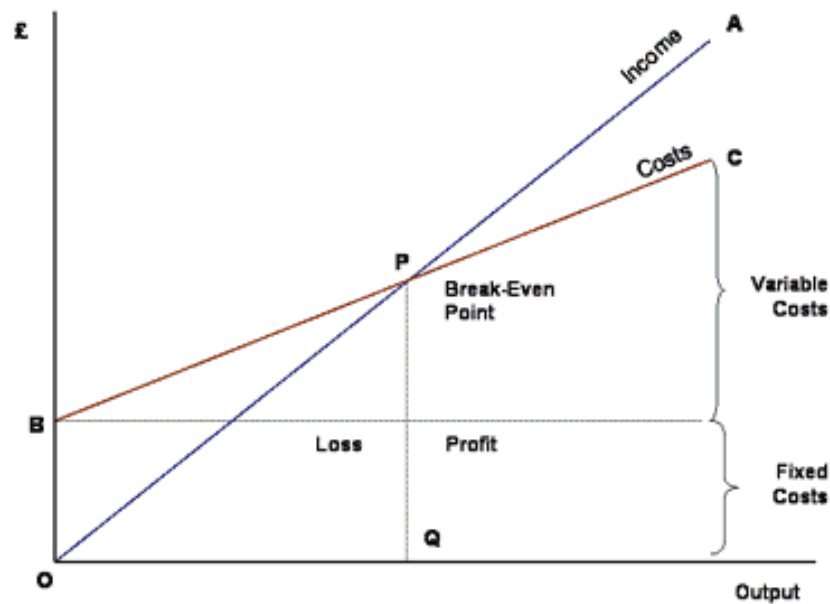
$$\text{Margin of Safety (in units)} = \text{Actual sale units} - \text{Break-even sales units}$$

$$\text{Margin of safety (in \%)} = \frac{\text{Margin of safety}}{\text{Total Sales}} \times 100$$

$$\text{Margin of Safety (in \%)} = 100\% - \text{Break-even Sales \%}$$

$$\text{Margin of safety (in ₹)} = \frac{\text{Profit}}{\text{P/V Ratio}}$$

$$\text{Margin of safety (in units)} = \frac{\text{Profit}}{\text{Contribution per unit}}$$



(Total FC - Unavoidable FC)

$$8. \text{ Shut down point} = \frac{\text{Avoidable fixed cost}}{\text{Contribution per unit (or) PV Ratio}}$$

eg: FC P.V. of $\angle Y = ₹.2$
 units of $\angle Y = 10000$
 units of $\angle Y = 12000$ ✓
 Total FC = 2×10000
 = 20000

9. Points to Remember (PTRs)

(A) If fixed cost per unit is given then multiply it with the level of units at which such fixed cost per unit was computed.

(B) Apply price effect of Total FC and never apply on FC per unit

Question – 1

MNP Ltd sold 2,75,000 units of its product at ₹375 per unit. Variable costs are ₹175 per unit (manufacturing costs of ₹140 and selling cost ₹35 per unit). Fixed costs are incurred uniformly throughout the year and amount to ₹3,50,00,000 (including depreciation of ₹1,50,00,000). there are no beginning or ending inventories.

Required:

- Compute breakeven sales level quantity and cash breakeven sales level quantity.
- Compute the P/V ratio. → $\frac{(375-175)}{375} \times 100 = 53.33\%$
- Compute the number of units that must be sold to earn an income (EBIT) of ₹25,00,000.
- Compute the sales level achieve an after-tax income (PAT) of ₹25,00,000. Assume 40% corporate Income Tax rate.

Handwritten calculations:

- $\frac{FC}{\text{Cont. P.V.}} = \frac{3500}{(375-175)} = 1.758$
- $\frac{\text{Calc FC}}{\text{Cont. P.V.}} = \frac{(3500 - 1500)}{(375-175)} = \frac{2000}{200} = 10$
- $\frac{(3500 + 2500)}{(375-175)} = 1,87,500$
- $\frac{(3500 + 4166667)}{(375-175)} = 195833 \text{ units}$
- $\frac{2500}{\text{G.P.}} = 41.66.667$

Solution

(i) Contribution = ₹375 - ₹175 = ₹200 per unit.

$$\text{Break even Sales Quantity} = \frac{\text{Fixed cost}}{\text{Contribution margin per unit}} = \frac{\text{₹ } 3,50,00,000}{\text{₹ } 200} = 1,75,000 \text{ units}$$

$$\text{Cash Break even Sales Qty} = \frac{\text{Cash Fixed Cost}}{\text{Contribution margin per unit}} = \frac{\text{₹ } 2,00,00,000}{\text{₹ } 200} = 1,00,000 \text{ units.}$$

$$(ii) \text{ P/V ratio} = \frac{\text{Contribution/unit}}{\text{Selling Price/unit}} \times 100 = \frac{\text{₹ } 200}{\text{₹ } 375} \times 100 = 53.33\%$$

(iii) No. of units that must be sold to earn an Income (EBIT) of ₹ 25,00,000

$$\frac{\text{Fixed cost} + \text{Desired EBIT level}}{\text{Contribution margin per unit}} = \frac{3,50,00,000 + 25,00,000}{200} = 1,87,500 \text{ units}$$

(iv) After Tax Income (PAT) = ₹25,00,000

Tax rate = 40%

$$\text{Desired level of Profit before tax} = \frac{\text{₹ } 25,00,000}{60} \times 100 = \text{₹ } 41,66,667$$

$$\text{Estimate Sales Level} = \frac{\text{Fixed Cost} + \text{Desired Profit}}{\text{P/V ratio}}$$

$$\begin{aligned} \text{Or, } & \left(\frac{\text{Fixed Cost} + \text{Desired Profit}}{\text{Contribution per unit}} \times \text{Selling Price per unit} \right) \\ & = \frac{\text{₹ } 3,50,00,000 + \text{₹ } 41,66,667}{53.33\%} = \text{₹ } 7,34,42,091 \end{aligned}$$

Question – 2

AZ company has prepared its budget for the production of 2,00,000 units. The variable cost per unit is ₹ 16 and fixed cost is ₹ 4 per unit. The company fixes its selling price to fetch a profit of 20% on total cost.

You are required to calculate:

(i) Present break-even sales (in ₹ and in quantity)

(ii) Present profit-volume ratio $\frac{(24-16)}{24} \times 100 = 33.3333\%$

(iii) Revised break-even sales in Rs and the revised profit-volume ratio, if it reduces its selling price by 10%. $24 - 10\% = 21.6$

(iv) What would be revised sales in quantity and the amount, if a company desires a profit increase of 20% more than the budgeted profit and selling price is reduced by 10% as above in point (iii).

$$\text{Req. Pbf} = 20 \times (24 - 20) + 20\% = 9.60$$

$$\text{Req. Sales (Rs)} = \frac{(80 + 9.60)}{25.93\%} = \checkmark$$

$$\text{Rev. Pv} = \frac{(21.6 - 16)}{21.6} \times 100 = 25.93\%$$

$$\text{Rev. Bep} = \frac{(20 \times 24)}{25.93\%} = 30.858$$

Solution

- (i) Present Fixed cost = $4 \times 2,00,000 = ₹ 8,00,000$
 Present Profit = Total cost $\times 20\% = (16 + 4) \times 20\% = ₹ 4$
 Present Selling price = Cost + Profit = $(16 + 4) + 4 = ₹ 24$
 Contribution = Selling price – Variable cost = $24 - 16 = ₹ 8$
 Present Break-even sales units = $\frac{\text{Fixed cost}}{\text{Contribution per unit}} = \frac{8,00,000}{8} = 1,00,000 \text{ units}$
 Present Break-even sales value = $1,00,000 \times 24 = ₹ 24,00,000$
- (ii) Present profit-volume ratio = $\frac{\text{Contribution}}{\text{Selling price}} \times 100 = \frac{8}{24} \times 100 = 8.33\%$
- (iii) New Selling price per unit = $24 - 10\% = ₹ 21.60$
 New contribution per unit = $21.60 - 16 = ₹ 5.60$
 Revised PV ratio = $\frac{5.60}{21.60} \times 100 = 25.93\%$
 Revised break-even sales = $\frac{8,00,000}{25.93\%} = ₹ 30,85,229$
- (iv) Required profit = Existing profit $\times 120\% = (4 \times 2,00,000) \times 120\% = ₹ 9,60,000$
 Required sales quantity = $\frac{\text{Required profit} + \text{Fixed cost}}{\text{Contribution per unit}} = \frac{9,60,000 + 8,00,000}{5.60} = 3,14,286 \text{ units}$
 Required sales value = $3,14,286 \times 21.60 = ₹ 67,88,578$

Question – 3

You are given the following data:

	<u>Sales</u>	<u>Profit</u>
✓ Year 2020-21 →	₹1,20,000	₹8,000
✓ Year 2021-22 →	₹1,40,000	₹13,000

Find out:

- (a) PV ratio → $\frac{(13,000 - 8,000)}{(1,40,000 - 1,20,000)} = 25\%$
- (b) B.E. point → $\frac{22,000}{25\%} = 88,000$
- (c) Profit when sales are ₹1,80,000 → $(1,80,000 \times 25\%) - 22,000 = 23,000$
- (d) Sales required to earn a profit of ₹12,000 → $\frac{(12,000 + 22,000)}{25\%} = 1,36,000$
- (e) Margin of safety in year 2021-22.
 → $1,40,000 - 88,000 = 52,000$ or $\frac{13,000}{25\%} = 52,000$

Solution

- (a) P/V Ratio = $\frac{\text{Change in Profit}}{\text{Change in Sales}} \times 100 = \frac{13,000 - 8,000}{1,40,000 - 1,20,000} \times 100 = \frac{5,000}{20,000} \times 100 = 25\%$
- (b) Contribution for Year 2021-22 = $1,20,000 \times 25\%$
 Fixed cost + Profit = 30,000
 Fixed cost + 8,000 = 30,000
 Fixed cost = ₹22,000
 Break-even sales = $\frac{\text{Fixed Cost}}{\text{P/V Ratio}} = \frac{22,000}{25\%} = ₹88,000$
- (c) Profit = Contribution – Fixed cost = $(1,80,000 \times 25\%) - 22,000 = ₹23,000$

$$(d) \text{ Desired sales (in ₹)} = \frac{\text{Fixed Cost} + \text{Desired Profit}}{\text{P/V Ratio}} = \frac{22,000 + 12,000}{25\%} = ₹1,36,000$$

$$(e) \text{ Margin of Safety} = \text{Actual sale} - \text{Break-even sales} = 1,40,000 - 88,000 = ₹52,000$$

Question – 4

Mr. X has ₹2,00,000 investments in his business firm. He wants a 15% return on his money. From an analysis of recent cost figures, he finds that his variable cost of operating is 60% of sales, his fixed costs are ₹80,000 per year. Show computations to Answer the following questions:

- What sales volume must be obtained to break even? $\rightarrow \frac{80,000}{40\%} = 2l$
- What sales volume must be obtained to get 15 per cent return on investment? $\rightarrow \frac{80,000 + (2l \times 15\%)}{40\%} = 2.75l$
- Mr. X estimates that even if he closed the doors of his business, he would incur ₹25,000 as expenses per year. At what sales would be better off by locking his business up? $\rightarrow \frac{(80,000 - 25,000)}{40\%} = 1.275l$

Solution

$$\text{P/V Ratio} = 100 - 60\% = 40\%$$

$$(i) \text{ Break-even point} = \frac{\text{Fixed Cost}}{\text{PV Ratio}} = \frac{80,000}{40\%} = ₹2,00,000$$

$$(ii) \text{ Required return} = ₹2,00,000 \times 15\% = ₹30,000$$

$$\text{Required sales} = \frac{\text{Fixed Cost} + \text{Required Profit}}{\text{PV Ratio}} = \frac{80,000 + 30,000}{40\%} = ₹2,75,000$$

$$(iii) \text{ Shut Down Point or Minimum Sales} = \frac{\text{Avoidable FC}}{\text{PV Ratio}} = \frac{80,000 - 25,000}{40\%} = ₹1,37,500$$

Thus, Mr. X should shut down the business if the sales are less than ₹1,37,500.

Question – 5

R Ltd. produces and sells 60,000 units of product 'AN', at its Noida Plant. The selling price of the product is ₹15 per unit. The variable cost is 80% of selling price per unit. Fixed cost during this period is ₹4,20,000. The company is continuously suffering losses, and management plans to shut down the Noida Plant. The fixed cost is expected to be reduced by ₹2,50,000. Additional costs of plant shut down are expected at ₹25,000. You are required to comment on:

- Whether the Noida plant be shut down?
- Find the shut-down point in units. $\rightarrow \frac{2.50l - 25,000}{3} = 75,000$

Solution**(i) Statement of Profit**

Particulars	If plant is continued (₹)	If plant is shut down (₹)
Selling price	15 per unit	-
Less: Variable cost	12 per unit	-
Contribution	3 per unit	-
Capacity	60,000 units	-
Total contribution	$60,000 \times 3 = 1,80,000$	-
Less: Fixed cost	4,20,000	1,70,000
Less: Additional fixed cost	-	25,000
Loss	(2,40,000)	(1,95,000)

Avoid FC

Since the loss of Noida plant exceeds shut down cost, it is better to shut down the plant.

$$(ii) \text{ Shut down point} = \frac{\text{Total fixed cost} - \text{Shut down cost}}{\text{Contribution per unit}} = \frac{4,20,000 - 1,95,000}{3} = 75,000 \text{ units}$$

Question – 6

SK Ltd. has furnished the following data for the two years:

		<u>2017</u>	<u>2018</u>
Sales	→	₹ 8,00,000	?
Profit/volume ratio (P/V Ratio)	→	VC = 50% 50%	VC = 62.5% 37.5%
Margin of safety sales as % of total sales	→	40%	BSS = 78.125% 21.875%

There has been substantial savings in the fixed cost in the year 2018 due to the restructuring process. The company could maintain its sales quantity level of 2017 in 2018 by reducing the selling price. You are required to calculate the following:

- (a) Sales for 2018 in ₹ - 6.40l
- (b) Break-even sales for 2018 in ₹ - 6.40l
- (c) Fixed cost for 2018
- Handwritten notes: $VC \text{ for yr. 2017} = VC \text{ for yr. 2018}$
 $8l \times 50\% = \text{Sales} \times 62.5\%$
 $\text{Sales} = \frac{4l}{62.5\%} = 6.40l$
 $BSS = \frac{FC}{P/V} \Rightarrow 5l = \frac{FC}{37.5\%} \Rightarrow FC = 1.875l$

Solution**(a) Year 2017**

$$\text{Variable cost ratio} = 100 - \text{P/V Ratio} = 100 - 50\% = 50\%$$

$$\text{Variable cost} = \text{Sales} \times \text{Variable cost ratio} = 8,00,000 \times 50\% = ₹ 4,00,000$$

Year 2018

Since there is no change in sales quantity and no information has been provided for change in variable cost per unit.

$$\therefore \text{Variable cost of Year 2018} = \text{Variable Cost of Year 2017} = ₹ 4,00,000$$

$$\text{Variable cost ratio} = 100 - \text{P/V Ratio} = 100 - 37.5\% = 62.5\%$$

$$\frac{\text{Variable cost}}{\text{Sales}} = 62.5\%$$

$$\text{Sales} = \text{Variable cost} \div 62.5\%$$

$$\text{Sales} = 4,00,000 \div 62.5\% = ₹ 6,40,000$$

$$(b) \text{ Breakeven sales (in \%)} = 100 - \text{Margin of Safety (in \%)} = 100 - 21.875\% = 78.125\%$$

$$\text{Breakeven sales} = 6,40,000 \times 78.125\% = ₹ 5,00,000$$

$$(c) \text{ Breakeven sales} = \frac{\text{Fixed Cost}}{\text{P/V Ratio}}$$

$$\text{Fixed cost} = \text{Breakeven sales} \times \text{P/V Ratio} = 5,00,000 \times 37.5\% = ₹ 1,87,500$$

Question – 7

Axion Industries is a heavy industrial gear manufacturing company with a manufacturing setup based in Pune. Mr. Andrew, the CFO of the company furnished the following information to Mr. Joe who heads the Finance department.
 For FY 2024-25:
 $\text{Var. OH P.U.} = \frac{28,500 - 25,000}{50,000 - 40,000} = 0.35$
 $\text{Fixed} = 28,500 - (50,000 \times 0.35) = 11,000$

Particulars	Amount ₹(in crores)
Total Sales →	1,00,000
Raw material cost (✓) →	50,000
Direct wages (✓) →	15,000
Fixed & variable overheads →	25,000
Profit →	10,000
Total Number of Units sold →	40,000 units

$$\begin{aligned} \text{Cont.} &= 12 - 0.50 - (0.15 \times \frac{120}{100}) \\ &= 0.45 \end{aligned}$$

The market being very competitive and with the raw materials rates rising, Mr. Andrew raises his concern with Mr. Joe where he expects in the next financial year 2025-26 workers' wages to rise by 20%, fixed costs component to decrease by ₹500 crores. Total fixed & variable overhead however is to be ₹28,500 crores. Total fixed & variable overhead however is to be ₹28,500 crores. The total number of units expected to be sold would be 50,000.

Required:

Calculate the minimum number of units to be sold to sustain the same per unit profit in the financial year 2025-26 also. (Ignore further effects on Fixed costs.)

$$\frac{10,000}{40,000} = 0.25$$

$$\begin{aligned} \text{Sales} &= \frac{\text{FC} + \text{Pbt}}{\text{Gross P.U.}} \\ y &= \frac{(11,000 - 500) + (0.25)(12)}{0.45} \\ y &= 52,500 \end{aligned}$$

Solution

Calculation of Variable Overhead Per Unit & Fixed Cost (FY 2024-25)

$$\text{Variable cost per unit} = \frac{\text{Difference in total cost}}{\text{Difference in units}} = \frac{28,500 \text{ crores} - 25,000 \text{ crores}}{50,000 \text{ units} - 40,000 \text{ units}} = ₹0.35 \text{ crore per unit}$$

$$\text{Fixed cost} = ₹25,000 \text{ crore} - (40,000 \times 0.35) = ₹11,000 \text{ crore}$$

$$\text{Revised fixed cost (FY 2025-26)} = ₹10,500 \text{ crore}$$

$$\text{Desired profit per unit} = ₹10,000 \text{ crores} \div 40,000 \text{ units} = ₹0.25 \text{ crore per unit}$$

Calculation of Contribution per unit

Particulars	Per unit (₹in crore)
Selling price per unit	2.50
Material cost	1.25
Labour (0.375 × 120%)	0.45
Variable Overheads	0.35
Contribution per unit	2.05
	0.45

$$\text{Let units to be sold} = y$$

$$\text{Profit} = \text{Contribution} - \text{Fixed cost}$$

$$0.25y = 0.45y - 10,500 \text{ crore}$$

$$y = 52,500 \text{ units}$$

Thus, minimum no. of units to be sold to sustain same profit per unit = 52,500 units

$$SP = \frac{\text{Sale}}{\text{Units}} = \frac{(6,39,000 \div 71\%)}{60,000} = 15$$

Question – 8

SK a zero sugar cold drink manufacturing Indian company, is planning to establish a subsidiary company in Nepal to produce coconut flavored juice. Based on the estimated annual sales of 60,000 bottles of the juice, cost studies produced the following estimates for the Nepalese subsidiary:

	Total Annual Costs (₹)	Percent of total annual cost which is variable	VC	FC
Material	2,70,000	100%	2,70,000	-
Labour	1,97,000	80%	1,57,600	39,400
Factory Overheads	1,20,000	60%	72,000	48,000
Administration Expenses	52,000	35%	18,200	33,800
			5,17,800	1,21,200

The Nepalese production will be sold by manufacturer's representative who will receive a commission of 9% of the sale price. No portion of the Indian office expenses is to be allocated to the Nepalese subsidiary. You are required to:

- Compute the sale price per bottle to enable the management to realize an estimated 20% profit on sale proceeds in Nepal
- Calculate the break-even point in rupees value sales and also in number of bottles for the Nepalese subsidiary on the assumption that the sale price is ₹ 14 per bottle.

$$BEP = \frac{1,21,200}{4.11} = 29,489$$

$$\begin{aligned} & \text{VC} = 5,17,800 \div 60,000 = 8.63 \\ & \text{Comm.} = 14 \times 9\% = 1.26 \\ & \text{Cont. P.V.} = 4.11 \end{aligned}$$

Solution**(i) Computation of Sale Price Per Bottle**

	(₹)
Variable Cost:	
Material	2,70,000
Labour (₹ 1,97,000 × 80%)	1,57,600
Factory Overheads (₹ 1,20,000 × 60%)	72,000
Administrative Overheads (₹ 52,000 × 35%)	18,200
Total (A)	✓ 5,17,800
Fixed Cost:	
Labour (₹ 1,97,000 × 20%)	39,400
Factory Overheads (₹ 1,20,000 × 40%)	48,000
Administrative Overheads (₹ 52,000 × 65%)	33,800
Total (B)	✓ 1,21,200
Total Cost without commission (A + B)	→ 6,39,000
Add: Commission (9% of ₹ 9,00,000)	81,000
Add: Profit (20% of ₹ 9,00,000)	1,80,000
Sales Proceeds (6,39,000 ÷ 71%) (C)	✓ 9,00,000
No. of bottles (D)	✓ 60,000
Selling price per bottle (C ÷ D)	15

(ii) Statement of calculation of break-even point

Particulars	Amount (₹)
Sale price per bottle	14.00
(-) Variable cost per bottle $\left(\frac{5,17,800}{60,000}\right)$	8.63
(-) commission per bottle $(14 \times 9\%)$	1.26
Contribution per bottle	4.11
Break-even point (in number of bottles)	$\frac{1,21,200}{4.11} = 29,489$
Break-even point (in sales value)	$29,489 \times 14 = 4,12,846$

Question – 9

SK Ltd. manufacture and sales its product S-9. The following figures have been collected from cost records of last year for the product S-9:

Elements of Cost	Variable Cost Portion	Fixed Cost
Direct Material →	30% of Cost of Goods Sold ₹70	-
Direct Labour →	15% of Cost of Goods Sold ₹72	-
Factory Overheads →	10% of Cost of Goods Sold ₹72	₹ 2,30,000
General & Administration Overheads →	2% of Cost of Goods Sold ₹72	₹ 71,000
Selling & Distribution Overhead →	4% of Cost of Sales ₹82	₹ 68,000

Last Year 5,000 units were sold at ₹ 185 per unit. From the given data find the followings:

(a) Break-even Sales (in rupees)

(b) Profit earned during last year

(c) Margin of safety (in %)

(d) Profit if the sales were 10% less than the actual sales.

(Assume that administration overheads are related with production activity)

$$\begin{aligned} \text{COGS} = y &= 0.3y + 0.15y + 0.1y + 2.30 + 0.02y + 0.71 \\ y &= 72 \\ \text{COGS} + \text{S\&DOH} &= \text{COS} \\ 72 + (0.04 \times \text{COS}) + 68000 &= \text{COS} \\ \text{COS} &= 82 \end{aligned}$$

Solution**Working Notes:**

$$\text{(i) } \underline{\text{COGS}} = \{(\text{DM} - 0.3\text{COGS}) + (\text{DL} - 0.15\text{COGS}) + (\text{FOH} - 0.10\text{COGS} + ₹ 2,30,000) + (\text{G\&AOH} - 0.02\text{COGS} + ₹ 71,000)\}$$

$$\text{COGS} = 0.57 \text{ COGS} + ₹ 3,01,000$$

$$\text{COGS} = \frac{3,01,000}{0.43} = ₹ 7,00,000$$

$$\text{(ii) } \text{COS} = \text{COGS} + (\text{S\&DOH} - 0.04\text{COS} + ₹ 68,000)$$

$$\text{COS} = ₹ 7,00,000 + (0.04 \text{ COS} + ₹ 68,000)$$

$$\text{COS} = \frac{7,68,000}{0.96} = ₹ 8,00,000$$

(iii) Calculation of Variable Costs:

$$\text{Direct Material} \quad (0.3 \times ₹ 7,00,000) \quad ₹ \quad 2,10,000$$

$$\text{Direct Labour} \quad (0.15 \times ₹ 7,00,000) \quad ₹ \quad 1,05,000$$

Factory Overhead	$(0.10 \times ₹ 7,00,000)$	₹	70,000
General & Administration OH	$(0.02 \times ₹ 7,00,000)$	₹	14,000
Selling & Distribution OH	$(0.04 \times ₹ 8,00,000)$	₹	32,000
		₹	4,31,000

(iv) Calculation of total Fixed Costs:

Factory Overhead-	₹	2,30,000
General & Administration OH	₹	71,000
Selling & Distribution OH	₹	68,000
	₹	3,69,000

(v) $P/V \text{ Ratio} = \frac{\text{Sales} - \text{Variable Cost}}{\text{Sales}} \times 100 = \frac{(185 \times 5,000) - 4,31,000}{185 \times 5,000} \times 100 = 53.41\%$

(a) $\text{Break-Even Sales} = \frac{\text{Fixed Costs}}{P/V \text{ ratio}} = \frac{3,69,000}{53.41\%} = ₹ 6,90,882$

(b) $\text{Profit earned during the last year} = (\text{Sales} - \text{Total Variable Costs}) - \text{Total Fixed Costs}$
 $= (₹ 9,25,000 - ₹ 4,31,000) - ₹ 3,69,000 = ₹ 1,25,000$

(c) $\text{Margin of Safety (\%)} = \frac{\text{Sales} - \text{Breakeven}}{\text{Sales}} \times 100 = \frac{9,25,000 - 6,90,882}{9,25,000} \times 100 = 25.31\%$

(d) $\text{Profit if sales were 10\% less than the actual sales}$
 $= 90\% (₹ 9,25,000 - ₹ 4,31,000) - ₹ 3,69,000 = ₹ 75,600$

10. Dual Selling price or Dual variable cost questions

- It will lead to generation of dual contribution per unit

Steps to solve

- Find both contribution per unit
- First calculate total contribution from 1st option which will be sold first.
- Recover the required value (FC or Profit etc.) from this and then calculate the balance required value.

Question – 10

A dairy product company manufacturing baby food with a shelf life of one year furnishes the following information:

- On 1st January, 2019, the company has an opening stock of 20,000 packets whose variable cost is ₹ 180 per packet.
- In 2018, production was 1,20,000 packets and the expected production in 2019 is 1,50,000 packets. Expected sales for 2019 is 1,60,000 packets.
- In 2018, fixed cost per unit was ₹ 60 and it is expected to increase by 10% in 2019. The variable cost is expected to increase by 25%. Selling price for 2019 has been fixed at ₹ 300 per packet.

You are required to calculate the Break-even volume in units for 2019.

Cont. P.V. upto 20000 units = $300 - 180 = 120$
 Cont. beyond 20000 units = $300 - 225 = 75$

$$FC = (60 \times 1.10) + 101 = 79.20$$

$$VC = 180 + 25\% = 225$$

$$\text{Total Cont.} = 120 \times 20000 = 240$$

$$\text{Bal. cont.} = 79.20 - 240 = 55.20$$

$$\text{Bal. unit} = \frac{55.20}{75} = 73600$$

$$BEP = 20000 + 73600 = 93600$$

Solution

Total fixed cost for year 2019 = $1,20,000 \times 60 \times 110\% = ₹ 79,20,000$

Contribution per unit upto first 20,000 units (Opening stock units) = $300 - 180 = ₹ 120$

Contribution per unit beyond 20,000 units (After sale of opening stock) = $300 - (180 \times 125\%) = ₹ 75$

Total contribution on first 20,000 units = $₹ 20,000 \times 120 = ₹ 24,00,000$

Thus, fixed cost recovered from initial sale of 20,000 units = ₹ 24,00,000

Balance fixed cost to be recovered = $₹ 79,20,000 - ₹ 24,00,000 = ₹ 55,20,000$

Units to be sold for recovery of additional fixed cost = $\frac{55,20,000}{75} = 73,600$ units

Thus, Break-even point = $20,000 + 73,600 = 93,600$ units

11. Composite or Overall BEP

This concept is used when company deals in multiple products.

Particulars	Prod. A	Prod. B	Prod. C	Total
Sales	- ✓	- ✓	- ✓	○
(-) Variable cost	- ✓	- ✓	- ✓	○
Contribution	-	-	-	○ ✓
(-) Fixed Cost				○
Profit				○

○ Overall Contribution per unit = $\frac{\text{Weighted average of contribution per unit}}{\text{Total Contribution}} \times \text{Total Units}$

○ Overall P/V Ratio = $\frac{\text{Weighted average of P/V Ratio}}{\text{Total Sales}} \times 100$

○ Overall Break-even Point (in units) = $\frac{\text{Fixed Cost}}{\text{Overall Contribution Per unit}}$

○ Overall Break-even Point (in ₹) = $\frac{\text{Fixed Cost}}{\text{Overall P/V Ratio}}$

For individual product distribute in sales mix Ratio

12. Activity level % at BES = $\frac{\text{Break-even sales}}{\text{Total Sales at 100% level}} \times 100$

40% of sales = $\frac{240}{40} \Rightarrow \text{Sales} = 1.50 \text{ units}$ PAT = $[(1.50 \times 40) - 240] (1 - 0.40) = 21.60$

Question – 11

SK Ltd. manufactures and sells a single product X whose selling price is ₹ 100 per unit and the variable cost is ₹ 60 per unit. Cont. = 40

(a) If the Fixed Costs for this year are ₹ 24,00,000 and the annual sales are at 60% margin of safety, Calculate the rate of return on sales, assuming an income tax level of 40%. $\frac{21.60}{(1.50 \times 100)} \times 100 = 14.40\%$

(b) For the next year, it is proposed to add another product line Y whose selling price would be ₹ 150 per unit and the variable cost ₹ 100 per unit. The total fixed costs are estimated at ₹ 28,00,000. Cont. = 50

The sales mix of X : Y would be 5 : 3. Compute the breakeven sales in units for both the products. BES = 40%

Overall Cont. P.V. = $(40)(\frac{5}{8}) + (50)(\frac{3}{8}) = 43.75$

$\frac{280}{43.75} = 64000$

X = $64000 \times \frac{5}{8} = 40000$
Y = $64000 \times \frac{3}{8} = 24000$

Solution

(a) Contribution per unit = Selling price – Variable cost = ₹ 100 – ₹ 60 = ₹ 40

$$\text{Break-even point} = \frac{\text{Fixed Cost}}{\text{Contribution per unit}} = \frac{24,00,000}{40} = \underline{60,000 \text{ units}}$$

Break-even sale (in %) = 100 – Margin of safety = 100 – 60% of sales = 40% of sales

$$\frac{\text{Break-even sales in units}}{\text{Actual sales units}} = \frac{40}{100}$$

$$\frac{60,000}{\text{Actual sale units}} = 0.40$$

Actual sale units = 1,50,000 units

Particulars	₹
Sales Value (1,50,000 × 100) →	✓ 1,50,00,000
Less: Variable cost (1,50,000 × 60) →	90,00,000
Contribution →	60,00,000
Less: Fixed cost →	24,00,000
Profit →	36,00,000
Less: Income tax @ 40% →	14,40,000
Net Return	21,60,000
Rate of net return on sales $\left(\frac{21,60,000}{1,50,00,000} \times 100 \right)$	14.40%

(b) Contribution per unit of Product X = 100 – 60 = ₹ 40

Contribution per unit of Product Y = 150 – 100 = ₹ 50

$$\text{Overall contribution per unit} = \left(40 \times \frac{5}{8} \right) + \left(50 \times \frac{3}{8} \right) = ₹ 43.75$$

$$\text{Overall Break-even point} = \frac{\text{Fixed cost}}{\text{Overall Contribution per unit}} = \frac{28,00,000}{43.75} = 64,000 \text{ units}$$

Break-even point of Product X = 64,000 × 5/8 = 40,000 units

Break-even point of Product Y = 64,000 × 3/8 = 24,000 units

Question – 12

Two manufacturing companies A and B are planning to merge. The details are as follows:

	A	B
Capacity utilization (%)	→ 90	→ 60
Sales (₹)	63,00,000	48,00,000
Variable Cost (₹)	39,60,000	22,50,000
Fixed Cost (₹)	13,00,000	15,00,000

Assuming that the proposal is implemented, calculate:

(i) Break-Even sales of the merged plant and the capacity utilization at that stage.

(ii) Profitability of the merged plant at 80% capacity utilization. → $(1502 \times 80\% \times 45.67\%) - 280$

(iii) Sales Turnover of the merged plant to earn a profit of ₹ 60,00,000.

$$\frac{(280 + 60)}{45.67\%} = \checkmark$$

Handwritten calculations for Question 12:

Capacity utilization: 90% (A) and 60% (B) → Total 150%

Sales: 63,00,000 (A) and 48,00,000 (B) → Total 1,11,00,000

Variable Cost: 39,60,000 (A) and 22,50,000 (B) → Total 62,10,000

Fixed Cost: 13,00,000 (A) and 15,00,000 (B) → Total 28,00,000

Contribution: 23,40,000 (A) and 25,50,000 (B) → Total 48,90,000

Break-Even Point: $\frac{28,00,000}{45.67\%} = 61,33,333$

Profitability at 80% capacity utilization: $(1,11,00,000 \times 80\% \times 45.67\%) - 28,00,000 = 28,00,000$

Sales Turnover to earn profit of ₹ 60,00,000: $\frac{(28,00,000 + 60,00,000)}{45.67\%} = 1,88,33,333$

- (iv) When the merged plant is working at a capacity to earn a profit of ₹ 60,00,000, what percentage of increase in selling price is required to sustain an increase of 5% in fixed overheads.

Solution

(i) **Statement of Profit** (₹ in lakhs)

Particulars	Plant A	Plant B	Total
Sales	63 ÷ 90% = 70	48 ÷ 60% = 80	150
(-) Variable Cost	39.6 ÷ 90% = 44	22.5 ÷ 60% = 37.50	81.50
Contribution	26	42.50	68.50
(-) Fixed Cost	13	15	28
Profit	13	27.50	40.50

$$\text{Overall P\&V Ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100 = \frac{68,50,000}{1,50,00,000} \times 100 = 45.67\%$$

$$\text{Overall Break-even point (in ₹)} = \frac{\text{Fixed Cost}}{\text{Overall P\&V Ratio}} = \frac{28,00,000}{45.67\%} = ₹ 61,30,939$$

$$\text{Break-even point capacity} = \frac{\text{Break-even sales}}{\text{Total Sales at 100\% level}} \times 100 = \frac{61,30,939}{1,50,00,000} \times 100 = 40.87\%$$

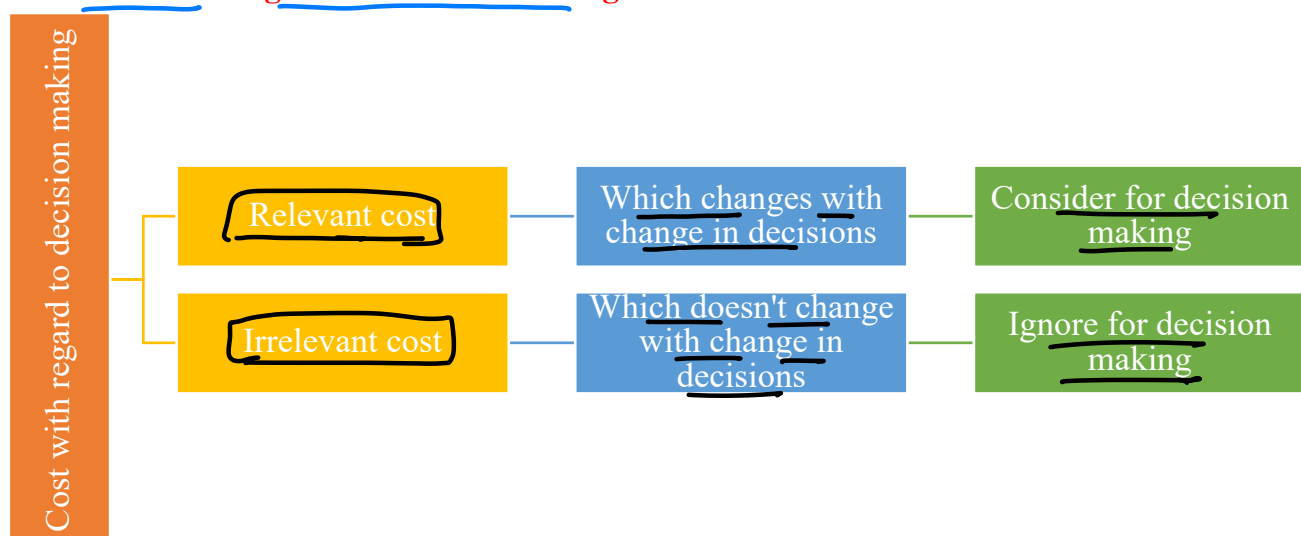
(ii) Sales at 80% level = 1,50,00,000 × 80% = ₹ 1,20,00,000

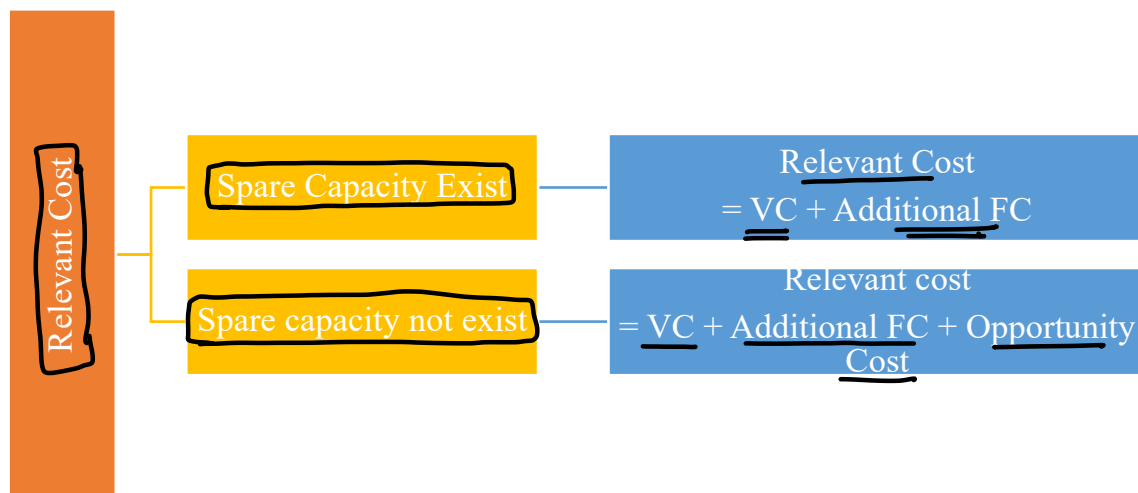
$$\text{Profit} = \text{Contribution} - \text{Fixed Cost} = (1,20,00,000 \times 45.67\%) - 28,00,000 = ₹ 26,80,400$$

$$\text{(iii) Desired Sales} = \frac{\text{Fixed Cost} + \text{Desired Profit}}{\text{Overall P\&V Ratio}} = \frac{28,00,000 + 60,00,000}{45.67\%} = ₹ 1,92,68,867$$

$$\text{(iv) Increase in fixed cost} = 28,00,000 \times 5\% = ₹ 1,40,000$$

$$\therefore \text{Percentage increase in selling price} = \frac{1,40,000}{1,92,68,867} \times 100 = 0.726\%$$

13. Cost with regard to decision making

14. Make vs Buy

Question – 13

XYZ Ltd. is a company involved in production and construction specialised equipment and machines on the demand of customers. The company received an order for construction of a specialised machine, it had nearly completed this job relating to construction of a specialised machine, when it discovered that the customer had gone out of business. At this stage, the position of the job was as under:

	(₹)
Original cost estimate →	27,50,000
Cost incurred so far (Paid) → 75%	24,80,000
Cost to be incurred → Rel.	✓ 3,70,000
Progress payment received from original customer (Paid) →	15,50,000 (88%)

After searches, a new customer for the machine has been found. He is interested to take the machine if certain modifications are carried out. The new customer wanted the machine in its original condition, but without its AI device and with certain other modifications. The costs of these additions and modifications are estimated as under:

Direct materials (at cost) →	₹1,05,000 ✓
Direct wages Dept.: X ✓	35 men days × 1000 = 35000
Dept.: Y ✓	55 men days × 1200 = 66000
Variable Overheads →	30% of Direct Wages in each Dept.
Delivery Costs →	₹15,500

3,70,000
 1,50,000
 35000
 66000
 (66000 × 3)
 198000
 30300
 15500
 80000
 (98500)
 193500
 150000
 45000
 1234800

Fixed overheads will be absorbed ⁽¹¹¹⁾ at 50% of direct wages in each department. The following additional information is available:

- (1) The direct materials required for the modification are in stock and if not used for modification of this order, they will be used in another job in place of materials that will now cost ₹1,50,000.
- (2) Department X is working normally and hence any engagement of labour will have to be paid at the direct wage rate of ₹1,000 per man day.
- (3) Department Y is extremely busy. Its direct wages rate is ₹1,200 per man day and it is currently yielding a contribution of ₹3 per rupee of direct wages.
- (4) Additional supervisory required for the modification cost ₹80,000. $105000 - (5 \times 1000) - (30\% \times 5000) = 98500$
- (5) The cost of the AI device that the new customer does not require is ₹1,35,000. If it is taken out, it can be used in another job in place of a different mechanism. The latter mechanism has otherwise to be bought for ₹1,05,000. The dismantling and removal of the control mechanism will take 5 man day in department X. $20 - (5 \times 1000) - (30\% \times 5000) = 193500$
- (6) If the conversion is not carried out, some of the materials in the original machine can be used in another contract in place of materials that would have cost ₹2,00,000. It would have taken 5 men days of work in department X to make them suitable for this purpose. The remaining materials will realize ₹1,50,000 as scrap. The drawings, which are included as part for the job can be sold for ₹45,000.

You are required to calculate the minimum price, which the company can afford to quote for the new customer as stated above.

Solution

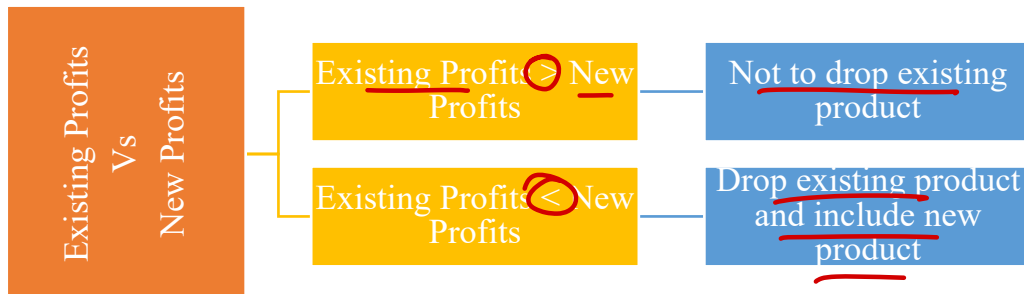
Statement of Minimum Price Which the Company Can Afford to Quote for the New Customer

	(₹)	(₹)
Cost to be incurred to bring the machine in its original condition		3,70,000
Direct Material (Replacement Value)		1,50,000
Direct Wages		
Dept. X: (35 men days × ₹ 1,000)	35,000	
Dept. Y: (55 men days × ₹ 1,200)	66,000	
Opportunity Cost of Contribution Lost by Dept. Y (₹66,000 × ₹3)	1,98,000	2,99,000
Variable Overheads [30% × (₹35,000 + ₹ 66,000)]		30,300
Delivery Costs		15,500
Additional Supervisory required for modification		80,000
Saving Due to Alternative Use of AI Device		

Bought Out Price	1,05,000	
Less: Dismantling & Removal Cost (5 men day × ₹1,000)	5,000	
Less: Variable Cost (30% × ₹ 5,000)	1,500	(98,500)
Net Loss on Material Cost Savings (W.N.)		1,93,500
Opportunity Cost of Remaining Materials which can be sold as scrap		1,50,000
Opportunity Cost of Sale of Drawings		45,000
Total Minimum Price which may be quoted		12,34,800

Working Note

	(₹)
Loss on Material Cost Saving of Machine	2,00,000
Less: Conversion Cost (5 men days × ₹1,000)	5,000
Less: Variable Cost (30% × ₹5,000)	1,500
Net Loss on Material Cost Saving of Machine	1,93,500

15. Dropping an existing product for new product**Question – 14**

SK Ltd. manufactures three different products and the following information has been collected from the books of accounts:

	Products		
	S	K	M
Sales Mix	→ 60% × 35% = 21%	→ 60% × 35% = 21%	→ 60% × 30% = 18%
Selling price	→ ₹ 300	→ ₹ 400	→ ₹ 200
Variable cost	→ $\frac{21}{30} \times 150 = ₹ 105$	→ ₹ 200	→ ₹ 120
Total Fixed costs			→ ₹ 18,00,000
Total Sales			→ ₹ 60,00,000

The company has currently under discussion, a proposal to discontinue the manufacture of product M and replace it with Product J, when the following results are anticipated:

	Products		
	<u>S</u>	<u>K</u>	<u>J</u>
Sales Mix →	50%	25%	25%
Selling price →	₹ 300	₹ 400	₹ 300
Variable cost →	₹ 150	₹ 200	₹ 150
Total Fixed costs	→ ₹ 18,00,000		
Total Sales	→ ₹ 64,00,000		

Required:

- Compute the PV ratio, total contribution, profit and Break-even sales for the existing product mix.
- Compute the PV ratio, total contribution, profit and Break-even sales for the proposed product mix.

Solution

(i) Computation of PV ratio

	Products		
	S	K	M
Selling price →	300	400	200
Less: Variable cost →	150	200	120
Contribution per unit	150	200	80
PV ratio	50%	50%	40%

$$\text{Overall PV Ratio} = \text{Weighted average PV ratio} = (50 \times 0.35) + (50 \times 0.35) + (40 \times 0.30) = 47\%$$

$$\text{Total Contribution} = \text{Sales} \times \text{Overall PV Ratio} = 60,00,000 \times 47\% = ₹ 28,20,000$$

$$\text{Total Profit} = \text{Contribution} - \text{Fixed Cost} = 28,20,000 - 18,00,000 = ₹ 10,20,000$$

$$\text{Break-even Sales} = \frac{\text{Fixed Cost}}{\text{Overall PV Ratio}} = \frac{18,00,000}{47\%} = ₹ 38,29,787$$

(ii) Computation of PV ratio

	Products		
	S	K	J
Selling price	300	400	300
Less: Variable cost	150	200	150
Contribution per unit	150	200	150
PV ratio →	50%	50%	50%

$$\text{Overall PV Ratio} = \text{Weighted average PV ratio} = (50 \times 0.50) + (50 \times 0.25) + (50 \times 0.25) = 50\%$$

$$\text{Total Contribution} = \text{Sales} \times \text{Overall PV Ratio} = 64,00,000 \times 50\% = ₹ 32,00,000$$

$$\text{Total Profit} = \text{Contribution} - \text{Fixed Cost} = 32,00,000 - 18,00,000 = ₹ 14,00,000$$

$$\text{Break-even Sales} = \frac{\text{Fixed Cost}}{\text{Overall PV Ratio}} = \frac{18,00,000}{50\%} = ₹ 36,00,000$$

Question – 15

JC Ltd. has a production capacity of 80,000 units per year. Presently a company produces 60,000 units. Its cost structure is as under:

Material Cost	₹6 per unit
Labour Cost	₹4 per unit
Variable overheads	₹2 per unit

Total fixed cost ₹3,00,000 per annum. Present selling price ₹20 per unit in the month of January, 2024 company received an offer from a Japanese client to supply 20,000 units at a price of ₹14 per unit with the additional shipping cost of ₹8,000.

Required:

- On the basis of changes in the profit, advice to the company, whether the offer should be accepted or not?
- Will your advice be different, if the customer is local one?
- If Japanese client offer for supply of 30,000 units to a price of ₹14 (part supply of order not accepted) and shipping cost treated as variable cost, analyze the impact on the profit of JC Ltd., if order accepted.

Solution

- (i) Statement Showing “Cost and Profit under Both Situation”

Particulars	Existing Production (60,000 units) (₹)	After Offer (80,000 units) (₹)
Sales		
Existing (60,000 × ₹20)	12,00,000	12,00,000
Offer (20,000 × ₹14)	-	2,80,000
Total Sales	12,00,000	14,80,000
Less: Direct Materials @ ₹6	3,60,000	4,80,000
Direct Labour @ ₹4	2,40,000	3,20,000
Variable Overheads @ ₹2	1,20,000	1,60,000
Contribution	4,80,000	5,20,000
Less: Additional Shipping cost	-	8,000
Less: Fixed Cost	3,00,000	3,00,000
Profit	1,80,000	2,12,000

Since the Profit has increased by ₹32,000, the proposal of the Japanese client should be accepted

- (ii) Yes, the advice will be different, if the customer is local one since the company is currently selling at ₹20 in local market and therefore, selling at discounted price of ₹14 may impact its local market.

- (iii) Statement Showing “Cost and Profit”

Particulars	After Offer (80,000 units) (₹)
Sales	
Existing (50,000 × ₹20)	→ 10,00,000

Offer ($30,000 \times ₹14$)	→ 4,20,000
Total Sales	→ 14,20,000
Less: Direct Materials @ ₹6	4,80,000
Direct Labour @ ₹4	3,20,000
Variable Overheads @ ₹2	1,60,000
Additional Shipping cost ($₹8,000/20,000$ units) $\times 30,000$ units	→ 12,000
Contribution	→ 4,48,000
Less: Fixed Cost	→ 3,00,000
Profit	1,48,000

If offer of Japanese client to supply 30,000 units at a price of ₹14 is accepted, the Profit will decrease by ₹32,000 from the current level.

Question – 16

The profit for the year of SK Ltd. works out to 12.5% of the capital employed and the relevant figures are as under:

Sales	→ ₹ 5,00,000	$-2.5\% - 1\% - 0.4\% - 0.5\% = 0.6\%$
Direct Materials	→ ₹ 2,50,000	$Sales = 50 \times \frac{110}{100} \times \frac{104}{100} = 5.72\%$
Direct Labour	→ ₹ 1,00,000	$VC = (2.5\% + 1\% + 0.4\%) \times \frac{110}{100} \times \frac{98}{100} = 4.2\%$
Variable Overheads	→ ₹ 40,000	$FC = 60,000 \times \frac{98}{100} = 58,800$
Capital Employed	→ ₹ 4,00,000	$\frac{92,780}{40} \times 100 = 23.19\% Profit = 92,780$

The new Sales Manager who has joined the company recently estimates for next year a profit of about 23% on capital employed, provided the volume of sales is increased by 10% and simultaneously there is an increase in selling price of 4% and an over cost reduction in all the elements of cost by 2%.

Required:

Find out by computing in details the cost and profit for the next year, whether the proposal of sales manager can be adopted.

Solution**Statement of Calculation of Fixed Overheads**

Particulars	Amount
Sales	5,00,000
(-) Profit ($4,00,000 \times 12.5\%$)	50,000
Total Cost	4,50,000
(-) Direct material	2,50,000
(-) Direct labour	1,00,000
(-) Variable overheads	40,000
Fixed overheads	60,000

Statement of Profit (Proposed Situation)

Particulars	Amount
Sales [$5,00,000 \times \frac{110}{100} \times \frac{104}{100}$]	5,72,000

Particulars	Amount
(-) Variable cost $[(2,50,000 + 1,00,000 + 40,000) \times \frac{110}{100} \times \frac{98}{100}]$	4,20,420
Contribution	1,51,580
(-) Fixed Overheads $(68,000 \times \frac{98}{100})$	58,800
Profit	92,780

$$\text{Profit as \% of capital employed} = \frac{92,780}{4,00,000} \times 100 = 23.195\%$$

Since the profit as % of capital employed is increasing, thus it is recommended to accept the proposal.

16. Key factor or limiting factor

It is the factor which is limited in its availability

Decision will be taken on the basis of contribution per unit of key factor

Key Factor

Basis of Decision

Sales (in units)	→	Contribution per unit
Sales (in ₹)	→	P/V Ratio
Material	→	Contribution <u>per unit of material</u>
Labour hour	→	Contribution <u>per labour hour</u>
Machine hour	→	Contribution <u>per machine hour</u>

Question – 17

Moon Ltd. produces products 'X', 'Y' and 'Z' and has decided to analyse its production mix in respect of these three products – 'X', 'Y' and 'Z'.

You have the following information:

	X	Y	Z
Direct materials (₹) per unit →	160	120	80
Variable overheads (₹) per unit →	8	20	12

X	Y	Z
312	400	240
168	140	92
72	160	108
72	180	40
6	10	5
12	10	8
I	II	III

Direct labour:

Departments: Rate per hour (₹) Hours per unit Hours per unit Hours per unit

		X	Y	Z
Department-A	4	6 = 24	10 = 40	5 = 20
Department-B	8	6 = 48 72	15 = 120 160	11 = 88 108

From the current budget, further details are as below:

	X	Y	Z
Annual <u>production</u> at present (in units) →	10,000	12,000	20,000
Estimated <u>selling price</u> per unit (₹) →	312	400	240
Sales department estimate of <u>possible sales</u> in the <u>coming year</u> (in units) →	12,000 ✓	16,000 ✓	24,000

There is a constraint on supply of labour in Department-A and its manpower cannot be increase beyond its present level. $\rightarrow (10000 \times 6) + (12000 \times 10) + (20000 \times 5) = 2.80 \text{ l}$

Required:

- Identify the best possible product mix of Moon Ltd.
- Calculate the total contribution from the best possible product.

X	12000	6	72000
Y	16000	10	160000
Z	48000	5	48000
			<u>280000</u>

Solution

Present supply of labour hours in Department-A

$$= (10,000 \times 6) + (12,000 \times 10) + (20,000 \times 5) = 2,80,000 \text{ labour hours}$$

Statement of Contribution

Particulars	X	Y	Z
Selling price per unit	312	400	240
(-) Direct material per unit	160	120	80
(-) Labour cost per unit			
Department A	4×6 = 24	4×10 = 40	4×5 = 20
Department B	8×6 = 48	8×15 = 120	8×11 = 88
(-) Variable overheads per unit	8	20	12
Contribution per unit	72	100	40
Labour hours per unit	6	10	5
Contribution per labour hour	12	10	8
Rank	I	II	III

Statement of Product Mix and Contribution

Product	Units	Labour hours per unit	Labour Hours consumed	Contribution
X	12,000	6	72,000	$72,000 \times 12 = 8,64,000 = 12000 \times 72$
Y	16,000	10	1,60,000	$1,60,000 \times 10 = 16,00,000 = 16000 \times 100$
Z	$48,000 \div 5 = 9,600$	5	(Bal. fig.) 48,000	$48,000 \times 8 = 3,84,000 = 9600 \times 40$
	37,600		2,80,000	28,48,000

find $\rightarrow \text{Roz} = 4000 + 3500 = 7500$
 Lily = $2000 + 1000 = 4000$

Question – 18

ABC Ltd. is a well-known company for producing baby care products. The company produces and sells two variants of organic shampoo for children: "Baby Rose" and "Baby Lily". The sales and cost data for both products are provided below:

Particulars	Baby Rose	Baby Lily
Current demand and Sales (Number of bottles)	4,000	3,000
Production Capacity (Number of bottles)	7,500	6,000
Selling Price per bottle (₹)	600	750
Variable Costs per bottle:		
- Direct Materials (₹20 Per litre)	160	200
- Other Variable Costs	270	350

Present mix = $(4000 \times 8) + (3000 \times 10) = 62000 \text{ Hr.}$

Bal. = $12 - 62000 = 28000 \rightarrow \text{Baby Roz} = (3500 \times 8) = 28000$

Rem. Bal. = 10000

Cost & Management Accounting

Max. Baby Roz = $\frac{28000}{8} = 4750$

Lily = $\frac{10000}{10} = 1000$

The fixed costs amount to ₹5,00,000 and ₹4,50,000 for Baby Rose and Baby Lily respectively. The Production Manager has informed that 1,00,000 litres of material is available for production. A dealer has approached the company and proposed to purchase both products at the existing selling prices, which are to be produced by utilizing the remaining unused material. However, he has insisted that all the bottles must be packed with eco-friendly packaging, which will result in an additional cost of ₹10 per bottle for the company. Presently, the company is not using eco-friendly material for packing of bottles.

Required:

Prepare a detailed statement showing the overall contribution and profit of the company after acceptance of the dealer's proposal.

Solution

Particulars	Baby Rose	Baby Lily	Total
	(₹)	(₹)	(₹)
Selling price per bottle	→ 600	750	—
Less: Direct Materials	→ 160	200	—
Other variable costs	→ 270	350	—
Contribution per bottle Before additional packaging	→ 170	200	—
Contribution per bottle per unit of raw material Before additional packaging	→ 21.25	20	
Ranking on the basis of Contribution per bottle per unit of raw material	I	II	

Particulars	Current Sales (WN2)	Additional Sales of Baby Rose (3,500 bottles)	Additional Sales of Baby Lily (1,000 bottles)	Total
	(₹)	(₹)	(₹)	(₹)
Selling price per bottle	—	600	750	—
Less: Direct Materials	—	160	200	—
Other variable costs	—	270	350	—
Additional packaging	—	10	10	—
Contribution per unit	—	160	190	—
Total Contribution	✓ 12,80,000	→ 5,60,000	→ 1,90,000	20,30,000
Less: Fixed Cost	✓ 9,50,000	—	—	9,50,000
Profit	3,30,000	5,60,000	1,90,000	10,80,000

WN1

	Baby Rose	Baby Lily
Raw Material used <u>per unit</u> of bottle (a)	8 litres ✓ (₹160/ ₹20)	10 litres ✓ (₹200/ ₹20)
Current Demand and Sales (b)	→ 4,000 bottles	3,000 bottles
Total Raw Material used (c = a × b)	→ 32,000 litres	30,000 litres

WN2

Statement showing the current contribution and profit of the company

Particulars	Baby Rose	Baby Lily	Total
	(₹)	(₹)	(₹)
Selling price per bottle	600	750	—
Less: Direct Materials	160	200	—
Other variable costs	270	350	—
Contribution per bottle Before additional packaging	170 ✓ × 4,000	200 ✓ × 3,000	—
Contribution per bottle per unit of raw material Before additional packaging	21.25	20	
Total Contribution Before additional packaging	6,80,000	6,00,000	12,80,000
Less: Fixed Cost	5,00,000	4,50,000	9,50,000
Profit	1,80,000	1,50,000	3,30,000

WN3

Raw Material available after current sales = 1,00,000 litres – 62,000 litres = 38,000 litres

Since the contribution per unit of Baby Rose is higher than Baby Lily, the company will produce and sale Baby Rose shampoo to the dealer.

Number of units that can be produced in 38,000 litres = 38,000 litres/8 litres = 4,750 bottles

However, the Production capacity of Baby Rose is 7,500 bottles, only 3,500 bottles can be produced.

Raw materials used in 3,500 bottles = 8 litres × 3,500 bottles = 28,000 litres

Remaining material = 10,000 litres

Number of Baby Lily that can be produced in 10,000 litres = 10,000 litres/10 litres = 1,000 bottles

17. Indifference Level

Level at which cost of two options will be equal

$$\text{Indifference level} = \frac{\text{Difference in Fixed cost}}{\text{Difference in variable cost per unit}}$$

OR Total cost of option (i) = Total cost of option (ii)
 $(VC1)(Q) + FC1 = (VC2)(Q) + FC2$
 Solve and find Q i.e. indifference level

Level	Recommendation
Actual quantity <u>></u> Indifference level	Select option having <u>variable cost per unit is low</u>
Actual quantity <u><</u> Indifference level	Select option where <u>fixed cost is low</u>
Actual quantity <u>=</u> Indifference level	Select <u>any option</u>

In case if there are three 3 options, then compute as follows:

- (a) Case 1 & 2
- (b) Case 2 & 3
- (c) Case 1 & 3

Question – 19

The following are cost data for three alternative ways of processing the clerical work for cases brought before the LC Court System:

	[A] <u>Manual</u> (₹)	[B] <u>Semi-Automatic</u> (₹)	[C] <u>Fully Automatic</u> (₹)
Monthly fixed costs:			
Occupancy	15,000	15,000	15,000
Maintenance contract	15,000 -	45,000 5,000	12,500 10,000
Equipment lease	-	25,000	1,00,000
Unit variable costs (per report):			
Supplies	240 40	140 80	40 20
Labour	200	60	20
	(5 hrs × ₹40)	(1 hr × ₹60)	(0.25 hr × ₹80)

Required:

- (a) Calculate cost indifference points. Interpret your results.
- (b) If the present case load is 600 cases and it is expected to go up to 850 cases in near future, select most appropriate on cost considerations?

Solution**(i) Cost Indifference Point**

$C_{A \& B} < 300 \rightarrow (A)$
 $C_{A \& B} = 300 \rightarrow (A \text{ or } B)$
 $300 < C_{A \& B} < 550$
 $C_{A \& B} = 550$

$550 < C_{B \& C} < 800$
 $C_{B \& C} = 800$
 $800 < C_{B \& C}$

	<u>A & B</u>	<u>A & C</u>	<u>B & C</u>
Differential fixed cost (₹) (A)	$45,000 - 15,000 = 30,000$	$1,25,000 - 15,000 = 1,10,000$	$1,25,000 - 45,000 = 80,000$
Differential variable cost (₹) (B)	$240 - 140 = 100$	$240 - 40 = 200$	$140 - 40 = 100$
Cost Indifference Point (A ÷ B)	300 cases	550 cases	800 cases

(ii) Interpretation of Results

At activity level below the indifference points, the alternative with lower fixed costs and higher variable costs should be used. At activity level above the indifference point alternative with higher fixed costs and lower variable costs should be used.

No. of Cases	Alternative to be Chosen
Cases ≤ 300	Alternative 'A'
$300 < \text{Cases} \leq 800$	Alternative 'B'
Cases ≥ 800	Alternative 'C'

18. Income statement under Marginal Costing

Particulars	Amount
Revenue (A)	-
Direct Material	-
Direct Labour	-
Direct expenses	-
Variable manufacturing overheads (Actual Basis)	-
Variable GFC/NFC/COP	-
Add: Opening stock of finished goods (Std. Cost P.u.)	-
Less: Closing stock of finished goods ($\frac{COP}{Prod} \times Q. St. units$)	-
Variable COGS	-
Add: Variable administration overheads	-
Add: Variable selling & distribution overheads	-
Variable COS (B)	-
Contribution (A - B)	-
Less: Fixed manufacturing overheads	-
Less: Fixed administration overheads	-
Less: Fixed selling & distribution overheads	-
Profit	-

19. Income statement under Absorption Costing

Particulars	Amount
Revenue (A)	-
Direct Material	-
Direct Labour	-
Direct expenses	-
Variable manufacturing overheads (Actual Basis)	-
Fixed manufacturing overheads (Recovery Basis)	-
GFC/NFC/COP	-
Add: Opening stock of finished goods (Std. Cost P.v.)	-
Less: Closing stock of finished goods $\left(\frac{\text{COP}}{\text{unit prod}} \times \text{Cl. St. units} \right)$	-
COGS	-
Add: Fixed & Variable administration overheads	-
Add: Fixed & Variable selling & distribution overheads	-
COS	-
Add: Under absorbed fixed manufacturing overheads	-
Less: Over absorbed fixed manufacturing overheads	-
Total Cost (B)	-
Profit/(loss) (A – B)	-

20. Profit of Marginal and Absorption differ due to difference in the stock values under both methods

Particulars	Amount
Profit as per Marginal Costing	-
(+) Opening stock under valued in Marginal	-
(-) Closing stock under valued in Marginal	-
Profit as per Absorption Costing	-

Question – 20

SK Ltd. has a production capacity of 2,00,000 units per year. Normal capacity utilization is reckoned as 90%. Standard variable production costs are ₹ 11 per unit. The fixed costs are ₹ 3,60,000 per year. Variable selling costs are ₹ 3 per unit and fixed selling costs are ₹ 2,70,000 per year. The unit selling price is ₹ 20.

$$\text{Sale} = \text{Op.} + \text{Prod.} - \text{Cl.}$$

$$1.502 = \text{Op.} + 1.602 - 0.202 \Rightarrow \text{Op.} = 10000$$

In the year just ended on 31st March, 2019, the production was 1,60,000 units and sales were 1,50,000 units. The closing inventory on 31st March was 20,000 units. The actual variable production costs for the year were ₹ 35,000 higher than the standard.

$$(1.602 \times 11) + 35000 = 17.952$$

(a) Calculate the profit for the year

- By absorption costing method and
- By marginal costing method

(b) Explain the difference in the profits

Solution

Working Note

Particulars	Year 2019
Opening stock	(Bal. fig.) 10,000
(+) Production	1,60,000
(-) Sales	1,50,000
Closing Stock	20,000

(a) Income Statement under Absorption Costing

Particulars	Amount
Sales (A)	$1,50,000 \times 20 = 30,00,000$ ✓
Variable Production Cost	$1,60,000 \times 11 = 17,60,000$ ✓
Under Recovered Variable Prod. Cost	35,000
Fixed Production	$\frac{3,60,000}{2,00,000 \times 90\%} \times 1,60,000 = 3,20,000$ ✓
GFC/NFC/COP	→ 21,15,000
(+) Op. Stock FG	$10,000 \times (11 + 2) = 1,30,000$ ✓
(-) Cl. Stock FG	$\frac{21,15,000}{1,60,000} \times 20,000 = 2,64,375$ ✓
COGS	19,80,625
(+) Variable Selling Cost	$1,50,000 \times 3 = 4,50,000$ ✓
(+) Fixed Selling Cost	→ 2,70,000
COS	27,00,625 ✓
(+) Under Recovered Fixed Prod. Cost	$3,60,000 - 3,20,000 = 40,000$ ✓
Total Cost (B)	→ 27,40,625
Profit (A – B)	2,59,375 ✓

Income Statement under Marginal Costing

Particulars	Year 2019
Sales (A)	$1,50,000 \times 20 = 30,00,000$ ✓
Variable Production Cost	$1,60,000 \times 11 = 17,60,000$ ✓
Under recovered variable Prod. Cost	35,000
Variable GFC/NFC/COP	17,95,000 ✓
(+) Op. Stock FG	$10,000 \times 11 = 1,10,000$ ✓
(-) Cl. Stock FG	$\frac{17,95,000}{1,60,000} \times 20,000 = 2,24,375$ ✓
Variable COGS (B)	→ 16,80,625
(+) Variable Selling cost	$1,50,000 \times 3 = 4,50,000$ ✓

Variable COS (B)	✓ 21,30,625
Contribution (A – B)	→ 8,69,375
(-) Fixed Production Cost	3,60,000
(-) Fixed Selling Cost	2,70,000
Profit	2,39,375

(b) The difference in profit is due to the valuation of stock in both methods.